C PROGRAMMING PROJECT PRESENTATION

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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

Project Title: [Password Vault and Credential Storage System]

Course: Java Programming Lab / Mini Project

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Department: Computer Science

Mentor / Guide: [Name & Designation]

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INTRODUCTION

Overview of the Password Vault System and its purpose

Problem background:

Users struggle to manage multiple passwords securely

Motivation:

Need for a secure and simple password storage system

Significance:

Helps users store and access credentials safely

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OBJECTIVE

Main goals of the project:

- To design and implement a Java-based password vault
- To provide secure login using a master password
- To store and retrieve credentials efficiently
- To apply object-oriented programming concepts

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PROBLEM STATEMENT

Current Issues:

- Users forget passwords frequently
- Passwords stored in unsafe ways
- High risk of security breaches

Solution:

Develop a secure password vault system using master password authentication

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SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i3 or higher
- RAM: 4 GB minimum
- Storage: 500 MB free space

Software Requirements:

- OS: Windows / Linux
- Java JDK
- IDE: Eclipse / IntelliJ / VS Code

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SYSTEM DESIGN

Structure and workflow of the system:

- User Registration (Master Password)
- Login Authentication
- Vault System (Store & Retrieve Data)
- Menu-driven Interface

Flow:

Login → Menu → Add/View Credentials → Exit

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MODULE DESCRIPTION

Authentication Module:

– Handles registration and login

Vault Module:

– Stores and retrieves credentials

User Interface Module:

– Displays menu and accepts input

Exit Module:

– Safely terminates program

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ALGORITHM / FLOWCHART

1. Start
2. Register master password
3. Login using password
4. Validate user
5. Display menu options
6. Perform selected operation
7. Repeat until exit
8. Stop

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IMPLEMENTATION

Techniques used:

- Java Classes (AuthService, VaultService)
- Loops and conditional statements
- Scanner for user input
- ArrayList for storing credentials

Includes authentication and vault management logic

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OUTPUT SCREENS

- Registration screen
- Login success / failure
- Add credential screen
- View stored credentials

(Insert screenshots here)

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RESULTS & DISCUSSION

- System successfully stores credentials
- Login authentication works correctly
- Easy to use and efficient

Challenges:

- No encryption implemented
- No database storage

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CONCLUSION & FUTURE ENHANCEMENT

Conclusion:

- Project achieved its objective
- Demonstrates Java OOP concepts

Future Enhancements:

- Add encryption (AES)
- Connect to database (MySQL)
- Develop GUI (JavaFX/Swing)
- Add password generator

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REFERENCES / ACKNOWLEDGEMENT

References:

- Oracle Java Documentation
- GeeksforGeeks
- W3Schools
- JavaTPoint

Acknowledgement:

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THANK YOU